TALENTRA DATABASE DESIGN HACKATHON

Milestone 8 — Relationship Models (B27–B29)

Status: Proposed relationship draft for team review. Cardinalities and optionality below are design proposals, not statements mandated by the docket.

1. Milestone scope

The docket asks for relationships across Administration, Organisation and account, and Gig marketplace. B27 covers user-to-role and role-to-permission; B28 covers organisation hierarchy and recruiter mapping; B29 covers assignments, handoff references and ratings.

2. B27 — Administration relationship model (Member A)

Relationship	Proposed cardinality	Design note
app_user → credential	1 to 0..N	A user may have multiple credential methods or historical credentials. If only one active credential per type is allowed, enforce that rule explicitly.
app_user ↔ role via user_role	M:N	Each mapping references one user and one role. Add uniqueness for the pair if duplicate assignments are not allowed.
role ↔ permission via role_permission	M:N	Each mapping references one role and one permission. Add uniqueness for the pair.
app_user → user_session	1 to 0..N	Sessions belong to one user; session revocation/end times define active status.
app_user → consent_record	1 to 0..N	Retain consent events/history as required; define consent purpose and version.
app_user → audit_event (actor)	1 to 0..N	Audit event may have a nullable actor for system-generated events if the policy permits it; append-only.
app_user → block_entry (blocker / blocked party)	1 to 0..N per role	Two FKs to app_user distinguish who blocks and who is blocked; prevent self-blocking if business rules require.

3. B28 — Organisation and account relationship model (Member B)

Relationship	Proposed cardinality	Design note
organisation → organisation_location	1 to 0..N	Each location belongs to one organisation; deletion policy must preserve required business history.
organisation → organisation_identifier	1 to 0..N	Identifiers carry country/type and protected text value; uniqueness scope must be defined (global or per country/type).
organisation → organisation_verification	1 to 0..N	Separate verification attempts/history from current organisation status if verification history is needed.
organisation ↔ recruiter via recruiter_organisation_mapping	M:N over time	Use valid_from/valid_to to model effective dates; define whether overlapping active mappings are allowed.
organisation → account	1 to 0..N (proposal)	Exact organisation/account boundary is not specified by the docket; confirm whether account is tenant, billing entity, or login grouping.
account ↔ app_user via account_admin	M:N (proposal)	Mapping indicates which users administer an account; confirm if a user may administer multiple accounts.
organisation → organisation (hierarchy)	parent 1 to 0..N children	Use a self-reference or closure table depending on query needs; prevent cycles if hierarchy must be a tree.

4. B29 — Gig marketplace relationship model (Member C)

Relationship	Proposed cardinality	Design note
app_user / organisation → gig_request	1 to 0..N requests	Choose requester_user_id and/or requester_organisation_id semantics; enforce exactly one or allowed combinations at the database level.
gig_worker_profile → gig_availability	1 to 0..N	Availability belongs to one worker. Define timezone source and overlap semantics.
gig_request → gig_application	1 to 0..N	A request can receive multiple worker applications.
gig_worker_profile → gig_application	1 to 0..N	A worker can apply to multiple requests; define duplicate-application rule.
gig_application → gig_assignment	0..1 assignment per accepted application (proposal)	Confirm whether assignment is created only after acceptance and whether an application can generate multiple assignments.
gig_assignment → gig_handoff	1 to 0..N (proposal)	Handoff references may be multiple events; external system owns payload or Talentra stores a snapshot—must be decided.
gig_assignment → gig_rating	0..N or 0..1 (decision)	Docket requires ratings but does not specify whether both parties rate or multiple ratings are allowed; agree rule before constraints.

5. Cross-module rules to confirm

• Entity names must match the team-approved Milestone 7 entity list. The names here follow the proposed draft and may need reconciliation.

• Master Data references (country, city, industry, skill, job role, status codes) should point to the owning Master Data module rather than duplicate lookup definitions.

• Administration owns user identity and credentials; Profile, Organisation and Gig should reference that identity rather than store duplicate credentials.

• Phase 1 Jobs & Applications should reference Organisation and Profile dependencies; do not treat Phase 2 entities as fully approved by this relationship draft.

• Document optionality and delete/update rules for each relationship; all foreign keys must later be database-enforced in Milestone 10.

• Where an entity is marked proposal or a cardinality is undecided, record the team decision before Milestone 9 columns are finalized.

6. Milestone 8 acceptance checklist

• B27 user-role and role-permission mappings are agreed and documented.

• B28 organisation hierarchy and effective-dated recruiter mapping are agreed and documented.

• B29 gig request/application/assignment/handoff/rating relationships are agreed and documented.

• Every relationship records cardinality, optionality, FK direction and ownership.

• Team records approval and open decisions before starting Milestone 9 (B30–B32).

Source vs proposal: The docket explicitly names the required relationship areas but does not define every table, cardinality, or optionality. Those details are marked as proposals for team validation.